

Specifications for an ABM of Brokerage in Matching Markets

This document specifies an agent-based model of brokered matching.

The model is a discrete-time simulation of a matching market with *agents* and a *broker*. Agents seeking pairwise matches either search on their own or outsource to the broker. All economic quantities (match output, fees, surplus) are in the same monetary units. All agents use heuristic decision rules. No agent holds beliefs about other agents' strategies.

The model specification covers agents (§1), the matching problem (§2), how agents learn to predict match quality (§3), match economics (§4), network structure and agent turnover (§5), search and broker access (§6), the outsourcing decision (§7), and period measures (§8). Baseline parameter values and reproducibility are documented in Part II (§9).

Part I. Model Specification

1. Agents and Time

The model runs for T discrete periods (default 500). It has N agents (default 1000), a single broker, and an undirected network G over the agents and broker.

Each agent i is characterized by:

- **Type** $\mathbf{x}_i \in \mathbb{R}^d$: a fixed vector of observable characteristics, drawn as specified below and used by the matching function (§2).
- **Current-period matches** M_i^t : active bilateral relationships involving i in period t . Each counterparty can appear at most once in M_i^t ; incoming accepted offers are not capped.
- **Active demand bound** K : the maximum number of active demands an agent can initiate in a period (default $K = 5$; §6).
- **Experience history** $\mathcal{H}_i^t = \{(\mathbf{x}_j, q_{ij})\}$: (other party's type, realized match output) pairs from all matches involving i (§3a).
- **Satisfaction indices** $s_{i,c}^t$: one scalar per search channel $c \in \{\text{self}, \text{broker}\}$, used in the outsourcing decision (§7).

Agent types

Type vectors are generated near a smooth one-dimensional curve on the unit sphere in $\mathbb{R}^d$. The curve is parameterized by a position $u \in [0, 1]$:

$$\mathbf{x}(u) = \frac{\tilde{\mathbf{x}}(u)}{\|\tilde{\mathbf{x}}(u)\|}, \quad \tilde{x}_\ell(u) = \begin{cases} \sin(2\pi\nu_\ell u + \psi_\ell) & \ell = 1, \dots, d_\gamma \\ 0 & \ell = d_\gamma + 1, \dots, d \end{cases}$$

where $\nu_\ell \sim \text{Unif}\{1, 2, 3, 4, 5\}$ are random integer frequencies, $\psi_\ell \sim \text{Unif}[0, 2\pi)$ are random phases, both drawn once per simulation, and $d_\gamma \leq d$ is the number of **active dimensions**. The remaining dimensions enter only through noise.

Each agent is drawn at a random position $u_i \sim \text{Unif}[0, 1]$ on the curve, then perturbed:

$$\mathbf{x}_i = \frac{\mathbf{x}(u_i) + \boldsymbol{\epsilon}_i}{\|\mathbf{x}(u_i) + \boldsymbol{\epsilon}_i\|}, \quad \boldsymbol{\epsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_x^2}{d} \mathbf{I}_d\right)$$

Noise is applied in all d dimensions, then the vector is re-normalized. The scale $\sigma_x/\sqrt{d}$ keeps the expected displacement from the curve approximately σ_x for any d .

d_γ controls type-space complexity by setting how many dimensions carry curve signal.

Broker

The broker is a permanent node in G and has no type vector. Its state is:

- **Experience history** $\mathcal{H}_b^t = \{(\mathbf{x}_i, \mathbf{x}_j, q_{ij})\}$: (party 1 type, party 2 type, realized match output) triples from mediated matches (ğ3c). The party labels have no directional meaning because the broker's pair features are symmetric.
- **Roster** Roster^t : the broker's standing access base (ğ6b).
- **Current clients** D^t : agents who outsource to the broker in period t (ğ6b).
- **Reputation** rep^t : the average broker satisfaction of current clients (ğ7).

2. The Matching Problem

Match output is symmetric and pair-level. The deterministic signal has three parts: agent general quality, pair interaction, and a regime-dependent gain applied to the interaction.

2a. Match quality

Let q_{ij} be the per-period value of the match between agents i and j , measured in monetary units:

$$q_{ij} = Q + f(\mathbf{x}_i, \mathbf{x}_j) + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

where $Q = 1.0$ is a constant offset, $\sigma_\varepsilon = 0.10$, and ε_{ij} is idiosyncratic match variation.

The deterministic signal $f : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is unknown to all agents and fixed for the duration of the simulation:

$$f(\mathbf{x}_i, \mathbf{x}_j) = \rho \cdot \frac{1}{2} [\mathbf{x}_i^\top \mathbf{c} + \mathbf{x}_j^\top \mathbf{c}] + (1 - \rho) \cdot g(\mathbf{x}_i, \mathbf{x}_j) \cdot \mathbf{x}_i^\top \mathbf{A} \mathbf{x}_j$$

where $\rho \in [0, 1]$ mixes general quality (ğ2b) and interaction (ğ2c).

2b. Agent general quality

$\mathbf{c}$ is an ideal type vector. Each agent’s general-quality score is $\mathbf{x}_i^\top \mathbf{c}$.

A pair contributes the average score $\frac{1}{2}(\mathbf{x}_i^\top \mathbf{c} + \mathbf{x}_j^\top \mathbf{c})$ to f . The ideal type $\mathbf{c} \in \mathbb{R}^d$ is drawn at initialization as a perturbation of a random point on the agent type curve:

$$u_c \sim \text{Unif}[0, 1], \quad \epsilon_c \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_x^2}{d} \mathbf{I}_d\right), \quad \mathbf{c} = \mathbf{x}(u_c) + \epsilon_c.$$

Unlike agent types, $\mathbf{c}$ is not re-normalized after the perturbation.

2c. Match-specific interaction

The interaction term is $g(\mathbf{x}_i, \mathbf{x}_j) \mathbf{x}_i^\top \mathbf{A} \mathbf{x}_j$, where g is a regime-dependent gain.

$\mathbf{A}$ is a symmetric positive definite interaction matrix. Matrix $\mathbf{A}$ gives a smooth bilinear interaction geometry, while $\mathbf{B}$ partitions pairs into gain regimes on the same type space. $\mathbf{A}$ and $\mathbf{B}$ are fixed for the duration of the simulation.

Base interaction. Draw $\mathbf{M}_A \in \mathbb{R}^{d \times d}$ with iid $\mathcal{N}(0, 1)$ entries and set

$$\mathbf{A} = \mathbf{M}_A^\top \mathbf{M}_A \cdot \frac{d}{\text{tr}(\mathbf{M}_A^\top \mathbf{M}_A)}.$$

Thus $\mathbf{A}$ is symmetric positive definite and $\text{tr}(\mathbf{A}) = d$.

For an isotropic unit vector $\mathbf{x}$, $E[\mathbf{x}\mathbf{x}^\top] = \mathbf{I}/d$, so $E[\mathbf{x}^\top \mathbf{A} \mathbf{x}] = \text{tr}(\mathbf{A})/d = 1$. This normalization fixes the average interaction scale.

Regime-dependent gain. A second symmetric matrix $\mathbf{B} \in \mathbb{R}^{d \times d}$ defines

$$g(\mathbf{x}_i, \mathbf{x}_j) = 1 + \delta \cdot \text{sign}(\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j)$$

where $\delta \in [0, 1]$ (default 0.5).

The regime depends only on the sign of $\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j$. Positive gives gain $1 + \delta$, negative gives $1 - \delta$, and zero gives gain 1. Because $\mathbf{B}$ is symmetric, the gain is symmetric in i and j .

The regime matrix $\mathbf{B}$ is constructed so the regime split is not a rescaling of the base interaction on the realized type distribution. Let

$$\Sigma_x = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^\top$$

be the uncentered second-moment matrix of the realized type vectors.

To draw the raw regime matrix, draw $\mathbf{Z} \in \mathbb{R}^{d \times d}$ with iid $\mathcal{N}(0, 1)$ entries, symmetrize it as $\mathbf{H} = (\mathbf{Z} + \mathbf{Z}^\top)/2$, and remove its average diagonal value:

$$\mathbf{B}_0 = \mathbf{H} - \frac{\text{tr}(\mathbf{H})}{d} \mathbf{I}_d.$$

This makes $\mathbf{B}_0$ symmetric with zero trace.

Under the weighted inner product defined as

$$\langle \mathbf{M}, \mathbf{N} \rangle_{\Sigma_x} = \text{tr}(\Sigma_x \mathbf{M} \Sigma_x \mathbf{N}),$$

remove the projection of $\mathbf{B}_0$ onto $\mathbf{A}$ and normalize:

$$\mathbf{B}_{\text{raw}} = \mathbf{B}_0 - \frac{\text{tr}(\Sigma_x \mathbf{B}_0 \Sigma_x \mathbf{A})}{\text{tr}(\Sigma_x \mathbf{A} \Sigma_x \mathbf{A})} \mathbf{A}, \quad \mathbf{B} = \frac{\mathbf{B}_{\text{raw}}}{\|\mathbf{B}_{\text{raw}}\|_F}.$$

This gives $\text{tr}(\Sigma_x \mathbf{B} \Sigma_x \mathbf{A}) = 0$ and $\|\mathbf{B}\|_F = 1$.

The weighted projection makes $\mathbf{B}$ orthogonal to $\mathbf{A}$ with respect to the realized type distribution, rather than with respect to unused directions in $\mathbb{R}^d$. In this context, orthogonality means

$$E\left[(\mathbf{x}_i^\top \mathbf{A} \mathbf{x}_j)(\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j)\right] = \text{tr}(\Sigma_x \mathbf{A} \Sigma_x \mathbf{B}) = 0,$$

where the expectation is over independent draws from the realized type distribution. Because agent types lie near a curve, this removes overlap between the two bilinear scores where agents actually occur.

$\mathbf{B}$ remains symmetric after projection and normalization.

Symmetry of $\mathbf{A}$ and $\mathbf{B}$ makes $f(\mathbf{x}_i, \mathbf{x}_j) = f(\mathbf{x}_j, \mathbf{x}_i)$.

3. Learning

Before searching for matches, agents and the broker may update their prediction models on their accumulated histories. Updated models are used to score candidates (§6). The broker is eligible to retrain every period. Agent i is eligible when $i \bmod 2 = t \bmod 2$, so each agent is eligible every other period. An eligible model retrains only when its history is nonempty and it has received new observations since its previous training call.

3a. Architecture and fitting

Both agents and the broker use the same architecture: a fully-connected network with one hidden layer, ReLU activations, and a single linear output:

$$\hat{q}(\mathbf{z}) = \mathbf{w}_2^\top \text{ReLU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) + b_2$$

where $\mathbf{z}$ is the input feature vector and the remaining symbols are network weights and biases. Agents receive raw partner type vectors, the broker receives pairs of them.

Training rule. Network weights minimize MSE with the Adam optimizer (learning rate η_{lr} , default 0.01) on the current training window. No explicit weight regularization is applied. At $t = 0$, each network sets $\mathbf{W}_1$ using He initialization, $\mathbf{b}_1 = \mathbf{0}$, $\mathbf{w}_2 = \mathbf{0}$, and $b_2 = Q$ (§2a). An untrained predictor therefore returns exactly Q for every input. Networks with seed observations then take E_{init} gradient steps (default 200) on their seed histories.

Subsequent updates warm-start from the previous weights and persistent Adam moment state. The number of gradient steps adapts to the ratio of new observations to total history:

$$E_t = \max\left(E_{\text{steps}}, \left\lceil E_{\text{init}} \cdot \frac{n_{\text{new}}}{n_{\text{total}}} \right\rceil\right)$$

where n_{new} is the number of observations added since that model’s previous training call, $n_{\text{total}} = |\mathcal{H}^t|$ is full history size, and E_{steps} (default 100) is the minimum per-call step count. For an agent, n_{new} can span multiple periods because agents train on an alternating schedule. Agents and the broker use the same step-count rule.

Initialization is period 0 for each learner. Its seed-history count is stored as the first period boundary. Each later boundary stores cumulative history at the end of a simulation period. A training call uses observations from the most recent $W_p = 40$ completed simulation periods. Before 40 simulation periods are available, the window also includes period 0. Once 40 are available, the period-0 seed history is excluded. If the selected history block exceeds $M_{\text{train}} = 2000$ observations, training deterministically keeps 2000 spaced observations across that block, including its oldest and newest observations. If the cap is one, it keeps the newest observation. The cap changes only the training sample, not the stored history or the period window.

3b. Agent i ’s model

History. $\mathcal{H}_i^t = \{(\mathbf{x}_j, q_{ij})\}_{m=1}^{n_i}$ records the other party’s type and realized match output from every match involving i .

Input. The agent’s network takes the partner’s type as input: $\mathbf{z} = \mathbf{x}_j$.

The hidden width is $h_A = 2d$, so the baseline uses 16 hidden units.

3c. Broker’s model

History. $\mathcal{H}_b^t = \{(\mathbf{x}_i, \mathbf{x}_j, q_{ij})\}_{m=1}^{n_b}$ records both parties’ types and realized match output from broker-mediated matches.

Input. The broker’s network takes a symmetric pair feature vector:

$$\psi(\mathbf{x}_i, \mathbf{x}_j) = \left[\mathbf{x}_i + \mathbf{x}_j; \text{vech}\left(\frac{\mathbf{x}_i \mathbf{x}_j^\top + \mathbf{x}_j \mathbf{x}_i^\top}{2}\right) \right].$$

The half-vectorization vech keeps the lower-triangular entries of the symmetric matrix, so the broker input dimension is $d_B = d + d(d+1)/2$, equal to 44 at $d = 8$.

The broker hidden width is $h_B = 8d$, so the baseline uses 64 hidden units.

Because $\psi(\mathbf{x}_i, \mathbf{x}_j) = \psi(\mathbf{x}_j, \mathbf{x}_i)$, each observed unordered pair contributes one training row.

The broker learns only from mediated matches. It does not observe self-search outcomes.

4. Match Economics

Matches use a common reservation threshold and channel-specific cost accounting.

At initialization, 10,000 random agent pairs define the calibration mean $\bar{q}_{\text{cal}} = E[q]$. This calibration quantity scales r , ϕ , and c_s .

4a. Outside options

All agents share a reservation threshold r , calibrated at initialization:

$$r = f_r \cdot \bar{q}_{\text{cal}}$$

where $\bar{q}_{\text{cal}}$ is the calibration mean and f_r is the reservation coefficient (default 0.60). Since ϕ and c_s do not depend on r (§4c), f_r may take any nonnegative value, including $f_r \geq 1$, and is swept as an analysis axis.

4b. Participation constraints

Demanders emit offers only above the reservation threshold. Self-search uses the demander evaluation $\hat{q}_i(\mathbf{x}_j) > r$. Brokered search uses the broker pair prediction $\hat{q}_b(\{i, j\}) > r$. A one-sided offer is accepted only if the receiver’s evaluation exceeds r ; reciprocal offers are accepted automatically (§6d).

4c. Search costs

The self-search cost and broker fee share one calibrated level:

$$\phi = c_s = \lambda_c \cdot \bar{q}_{\text{cal}}.$$

λ_c is the shared cost rate (default 0.05). The two symbols differ in payoff accounting:

- **Self-search** costs c_s on each requested self-search position, whether or not it is filled.
- **Brokerage** charges ϕ only on successful brokered placements.

5. Network Structure and Turnover

Agents interact through a single undirected network G that determines search opportunities and structural position.

5a. Network initialization

G is initialized as a Watts-Strogatz small-world graph on the N agents (Watts & Strogatz, 1998), with ring degree $k_G = 6$, rewiring probability $p_{\text{rewire}} = 0.1$, and random agent order on the ring.

The broker is added as a permanent node with no type vector. Broker-node edges are synchronized to standing roster members, current-period broker clients, and agents currently engaged in broker-channel matches (§6b).

5b. Match tie formation

Each realized match adds an undirected i - j edge if one does not already exist. Ties persist until either node exits.

5c. Agent turnover

Agents exit independently each period with probability η_{exit} (default 0.02). An exiting agent is removed everywhere. Its node slot is reused by an entrant with a fresh type drawn from the same type process (§1). The entrant receives $\lfloor k_G/2 \rfloor$ edges to existing agents sampled with probability proportional to $\exp(-\|\mathbf{x}_{i'} - \mathbf{x}_j\|^2)$.

6. Search

At the start of each period, each agent draws active relationship demand

$$d_i \sim \text{Binomial}(K, p_{\text{demand}}),$$

where K is the maximum number of active demands the agent can initiate in the period and p_{demand} defaults to 0.50. If $d_i > 0$, the agent chooses **one channel for the batch** (§7): self-search or broker.

6a. Self-search

Agent i 's self-search candidate pool has two components:

Neighbors. For a direct network neighbor j with prior pair-specific experience, agent i uses $\bar{q}_{ij} = \frac{1}{n_{ij}} \sum q_{ij}^{(m)}$, where n_{ij} is the number of times i and j have matched. If the graph edge exists but $n_{ij} = 0$, the agent instead uses $\hat{q}_i(\mathbf{x}_j)$. Initial histories seed one observation for every initial graph edge. Unobserved neighbor edges arise after turnover because entrants attach structurally before matching.

Strangers. Each self-searching demander i independently samples a pool U_i^t of $\min(n_{\text{strangers}}, N)$ agents uniformly without replacement, where $n_{\text{strangers}} = 10$ (default). The demander evaluates pool members that are not current neighbors using $\hat{q}_i(\mathbf{x}_j)$ (§3b).

Agent i orders feasible self-search candidates by these evaluations and emits up to d_i directed offers to candidates above r .

6b. Broker roster and access

The broker maintains a **roster** of agents it can propose as counterparties when mediating matches. In each period, it also observes the current-client set D^t of agents who outsourced. Broker access for the period is:

$$\mathcal{A}_b^t = \text{Roster}^t \cup D^t.$$

Current clients expand broker access for the period but do not become standing roster members.

Initialization. The roster is seeded with a target size

$$R^* = \lceil \alpha_R N \rceil,$$

where the broker roster share α_R defaults to 0.20. The model draws R^* agents uniformly at random (default 200 at $N = 1000$). The broker's history is seeded from existing roster-roster ties in the initial graph, capped at 100 observed ties.

Roster refresh. At the start of each period, each roster member independently exits with probability p_{roster} (default 0.02). If $\widetilde{\text{Roster}}^t$ is the post-churn roster,

$$\widetilde{\text{Roster}}^t = \{i \in \text{Roster}^{t-1} : u_i^t > p_{\text{roster}}\}, \quad u_i^t \stackrel{iid}{\sim} \text{Unif}[0, 1],$$

then the broker samples without replacement from $\{1, \dots, N\} \setminus \widetilde{\text{Roster}}^t$ until $|\text{Roster}^t| = R^*$, or until the population is exhausted. Broker-node edges are synchronized to the standing roster, current clients, and current broker-mediated matches (§5a).

6c. Broker-mediated search

For every unordered pair $\{i, j\}$ such that one side is a broker-client demander and the other side is in $\mathcal{A}_b^t$, the broker computes predicted match quality. For each broker-client demander i , it keeps the highest-valued accessible counterparties above r , up to i 's residual active broker demand. Directed broker offers are processed in the score order induced by the global pair ranking. If both sides are broker-client demanders with remaining active demand, the same unordered pair can generate reciprocal broker offers.

6d. Offer acceptance and match formation

All self-search and broker offers enter one shared market, grouped by unordered pair.

1. If pair $\{i, j\}$ contains reciprocal offers $i \rightarrow j$ and $j \rightarrow i$, the relationship is accepted automatically.
2. If pair $\{i, j\}$ contains one directed offer $i \rightarrow j$, receiver j evaluates i using $\bar{q}_{ji}$ for known partners and $\hat{q}_j(\mathbf{x}_i)$ otherwise. The offer is accepted iff this value exceeds r .
3. Each accepted unordered pair forms at most one current-period relationship.

At market finalization, for each accepted match: 1. Realized output is drawn: $q_{ij} = Q + f(\mathbf{x}_i, \mathbf{x}_j) + \varepsilon_{ij}$. 2. Both parties add the observation to their histories: i adds $(\mathbf{x}_j, q_{ij})$ to $\mathcal{H}_i$; j adds $(\mathbf{x}_i, q_{ij})$ to $\mathcal{H}_j$. 3. If either directed offer used the broker channel, the broker adds $(\mathbf{x}_i, \mathbf{x}_j, q_{ij})$ to $\mathcal{H}_b$ once. 4. An edge is added between i and j in G (if not already present).

A demander can have at most d_i outgoing offers, but can receive any number of acceptable incoming offers. Thus K is an active-demand bound, not a capacity constraint on receiving offers.

Before the next period begins, match lists are cleared. The next period begins with no active relationships.

7. The Outsourcing Decision

7a. Satisfaction tracking

Each agent i maintains a satisfaction index $s_{i,c}^t$ for each search channel $c \in \{\text{self}, \text{broker}\}$. For the chosen channel, the index updates once per active-demand period as an EWMA of realized match value net of search costs:

$$s_{i,c}^{t+1} = (1 - \omega) s_{i,c}^t + \omega \cdot \tilde{q}$$

where $\omega = 0.2$ and $\tilde{q}$ is the period input in the table. The denominator is requested demand d_i , so unfilled requests contribute zero output. A reciprocal relationship can update two channel-specific states, one for each directed offer.

Channel	Satisfaction input $\tilde{q}$
Self-search	$\frac{\sum q_{ij}}{d_i} - c_s$, summing over accepted directed offers made by i through self-search
Brokered	$\frac{\sum q_{ij} - \phi \cdot n_{i,\text{broker success}}}{d_i}$, summing over accepted directed offers made by i through the broker

Brokered failure gives $\tilde{q} = 0$, self-search failure gives $\tilde{q} = -c_s$.

$s_{i,\text{self}}^t$ covers all self-search, including known neighbors.

At initialization, self-satisfaction is the mean of seed match outcomes and broker-satisfaction is the broker's seed-data reputation (§7c). Fresh entrants set self-satisfaction to mean neighbor self-satisfaction and broker-satisfaction to current broker reputation.

The first broker choice, regardless of placement success, switches the agent from broker reputation to its own $s_{i,\text{broker}}^t$ (§7b).

7b. Decision rule

Each agent with $d_i > 0$ compares:

- **score_{self}** = $s_{i,\text{self}}^t$
- **score_{broker}** = $s_{i,\text{broker}}^t$ if the agent has tried the broker, otherwise rep_b^t .

The agent chooses the higher-score channel. Ties are resolved by fair coin flip.

7c. Broker reputation

$$\text{rep}_b^{t+1} = \begin{cases} \frac{1}{|D^t|} \sum_{i \in D^t} s_{i,\text{broker}}^{t+1} & \text{if } D^t \neq \emptyset \\ \text{rep}_b^t & \text{otherwise} \end{cases}$$

where D^t is the set of agents who outsourced to the broker this period. With current clients, reputation is their mean post-update broker satisfaction; with no clients, it is unchanged. Reputation is initialized from the mean broker seed outcome.

8. Period Measures

Period measures are evaluated after current-period matching, satisfaction, and reputation updates, but before agent entry/exit turnover. Agent decisions do not depend on these measures.

Network measures

Broker network-position measures use G , including the broker node (§5a). They are computed when $t \bmod \Delta_{\text{net}} = 0$ (default 20). Other period measures are recorded each period.

Betweenness centrality. Freeman betweenness (Freeman, 1977) for the broker node:

$$\text{BC}_b = \frac{1}{(n-1)(n-2)} \sum_{u \neq b} \sum_{v \neq u} \frac{\sigma_{uv}(b)}{\sigma_{uv}}$$

where $n = N+1$, σ_{uv} is the number of shortest paths from u to v , and $\sigma_{uv}(b)$ is the number passing through b . The double sum counts undirected pairs from both directions, so the denominator is $(n-1)(n-2)$ (Brandes, 2001, p. 9).

Burt’s constraint. Broker ego-network constraint (Burt, 1992):

$$\text{Constraint}_b = \sum_j \left(p_{bj} + \sum_{h \neq b, j} p_{bh} p_{hj} \right)^2$$

where $p_{bj} = 1/d_b$ and $p_{hj} = 1/d_h$ (Everett & Borgatti, 2020). The indirect term uses intermediary h ’s degree.

Effective size. Non-redundant broker contacts using the Borgatti (1997) binary-undirected simplification:

$$\text{ES}_b = d_b - \frac{2t_b}{d_b},$$

where d_b is the broker’s degree and t_b is the number of ties among the broker’s neighbors, excluding ties to the broker.

Prediction quality

Prediction quality is evaluated on holdout pairs and accepted directed offers:

Sample	Construction	Reported measures
Holdout pairs	Up to 100 agents with non-empty histories; for each sampled agent i , $\min(40, N-1)$ non-self partners j . Agent and broker predictions are evaluated against noiseless $Q + f(\mathbf{x}_i, \mathbf{x}_j)$. Both models use the same deterministic sample conditional on seed and period.	Per-agent R^2 , RMSE, bias, and Spearman rank correlation, averaged across sampled agents.

Sample	Construction	Reported measures
Selected offers	Accepted directed offers by channel, self-search or brokered. A reciprocal relationship can represent two directed decisions.	Channel-specific R^2 , RMSE, bias, and Spearman rank correlation.

The definitions are

$$R^2 = 1 - \frac{\text{MSE}}{\text{Var}_{\text{pop}}(q)}, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum (\hat{q} - q)^2}, \quad \text{Bias} = \frac{1}{n} \sum (\hat{q} - q).$$

Rank correlation is Spearman’s ρ_S between predicted and realized output. For selected-offer summaries, R^2 , bias, and rank correlation are undefined when $n < 5$ or $\text{Var}_{\text{pop}}(q) < \sigma_\varepsilon^2/6$, while RMSE is defined whenever $n > 0$. Holdout summaries use common valid-agent support for all four measures: an agent-level holdout set that fails the sample-size or variance condition is excluded from the R^2 , RMSE, bias, and rank-correlation means.

Other measures

Measure	Definition
Access vs. assessment	For each accepted broker-directed offer, record whether receiver j was a direct neighbor of sender i in G at the time of the offer. Non-neighbor receivers indicate access; neighbor receivers indicate assessment.
Match quality by channel	Average realized match output $\bar{q}_c^t$, where $c \in \{\text{self, brokered}\}$.
Mean channel satisfaction	Cross-agent means $N^{-1} \sum_i s_{i,\text{self}}^t$ and $N^{-1} \sum_i s_{i,\text{broker}}^t$.
Outsourcing rate	Outsourced requested positions divided by total requested positions.
Broker access size	$\ \mathcal{A}_b^t\ = \ \text{Roster}^t \cup D^t\ $, measured after outsourcing decisions form D^t and before entry/exit.
Counterparty concentration	Median and maximum current-period counterparties per agent.
Agent-node degree summaries	Mean, median, minimum, and maximum degree, excluding the broker node.

Part II. Baseline Values and Reproducibility

9. Baseline Values and Reproducibility

9a. Baseline parameter values

The table lists baseline values only. Entries are arranged in two columns; each column follows the document order, with the run length listed first.

Symbol	Baseline	Defined in	Symbol	Baseline	Defined in
T	500	g1	f_r	0.60	g4a
N	1000	g1	λ_c	0.05	g4c
d	8	g1	k_G	6	g5a
d_γ	8	g1	p_{rewire}	0.1	g5a
σ_x	0.5	g1	η_{exit}	0.02	g5c
Q	1.0	g2a	K	5	g6
ρ	0.50	g2a	p_{demand}	0.50	g6
δ	0.5	g2c	$n_{\text{strangers}}$	10	g6a
σ_ε	0.10	g2a	α_R	0.20	g6b
η_{lr}	0.01	g3a	p_{roster}	0.02	g6b
E_{init}	200	g3a	ω	0.2	g7a
E_{steps}	100	g3a	Δ_{net}	20	g8
W_p	40 periods	g3a	M_{train}	2000	g3a

9b. Reproducibility

All model-event randomness flows from a single integer seed. The seed determines type draws, G , matching function parameters ($\mathbf{c}$, $\mathbf{A}$, $\mathbf{B}$), broker seed roster, standing-roster churn and replenishment, and subsequent model events. Holdout samples are deterministic conditional on seed and period. Simulations are fully reproducible given parameters and seed.

References

- Brandes, U. (2001). A faster algorithm for betweenness centrality. *Journal of Mathematical Sociology*, 25(2), 163–177.
- Borgatti, S. P. (1997). Structural holes: Unpacking Burt’s redundancy measures. *Connections*, 20(1), 35–38.
- Burt, R. S. (1992). *Structural holes: The social structure of competition*. Harvard University Press.
- Everett, M. G., & Borgatti, S. P. (2020). Unpacking Burt’s constraint measure. *Social Networks*, 62, 50–57.
- Freeman, L. C. (1977). A set of measures of centrality based on betweenness. *Sociometry*, 40(1), 35–41.
- Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of ‘small-world’ networks. *Nature*, 393(6684), 440–442.